

Database Foundations

2-3 : Entities and Attributes

Practices

Exercise 1: Identify and draw entities as a beginning of an ERD

Overview

In this practice, you identify and add the entities for an Academic Database, or in other words a School Management System.

For your convenience, here is a summary of how the Academic Database (School Management System) works:

- A School/University has many Departments which offer courses to students in a given academic session.
- Each of these courses is taught by a faculty.
- Students enroll for different courses in an academic session.
- Besides the registration details, the parent information of the student also needs to be maintained by the University/School.
- The Department maintains the student's attendance details which would decide the eligibility of the student to take up the exams for that academic session.
- For each academic session, exams are conducted and the results are shared with the student within a stipulated period of time.
- The Department also maintains a log of the Faculty login and logout time for their reporting needs.

Tasks

- With the information provided above, identify and create the entities for the School Management System.

Exercise 2: Identify and add Attributes and corresponding Mandatory and Optional notation to ERD

Overview

In this practice, you identify the attributes and their associated optionality notation in the entities which you have created in Exercise 1.

Tasks

- Add the appropriate attributes as well as the optionality (*, °) to all the entities of the Academic Database.

